

ST LOUIS DOWNTOWN, IL, USA

WMO: 725314

Lat: 38.571N

Lon: 90.157W

Elev: 126

StdP: 99.82

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.8	-19.2	0.7	-11.8	-17.1	0.8	-9.5	11.7	-2.2	10.6	-1.5	3.8	310	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	35.1	24.8	33.7	24.5	32.5	24.0	26.5	32.5	25.7	31.6	25.1	30.7	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.3	29.8	24.0	19.1	28.9	23.1	18.1	28.0	83.3	32.8	79.7	31.9	76.8	30.6	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.3	8.3	7.4	DB	-17.4	37.5	3.1	1.7	-19.6	38.7	-21.4	39.7	-23.1	40.6	-25.3	41.8	(4)
(5)			WB	-17.8	28.1	2.9	1.0	-19.9	28.7	-21.6	29.3	-23.2	29.8	-25.3	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.3	2.3	7.9	14.0	19.4	23.9	25.9	25.0	20.9	14.5	7.8	2.2		
	DBStd	10.27	6.28	6.15	6.05	5.02	4.45	3.32	3.06	2.93	4.03	5.12	5.37	5.73		
	HDD10.0	1032	304	226	115	19	1	0	0	0	15	104	248			
	HDD18.3	2507	558	450	328	146	43	3	0	1	21	141	318	499		
	CDD10.0	2396	6	9	48	138	293	417	494	464	327	154	38	7		
	CDD18.3	829	0	0	3	16	77	171	236	206	98	23	1	0		
	CDH23.3	8404	0	2	29	163	669	1730	2587	2031	961	219	14	0		
	CDH26.7	3466	0	0	2	33	202	721	1202	874	372	57	1	0		
(14) Wind	WSAvg	3.1	3.8	3.9	3.9	3.9	3.0	2.6	2.3	2.1	2.2	2.8	3.3	3.5		
(15) Precipitation	PrecAvg	992	53	54	88	102	110	108	94	76	78	74	84	71		
	PrecMax	1557	229	133	213	264	328	334	272	169	248	314	253	298		
	PrecMin	596	2	6	28	25	20	11	3	2	0	5	11	17		
	PrecStd	203	40	31	39	58	62	69	55	40	55	49	52	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	22.0	26.3	29.8	32.6	35.3	37.9	36.6	34.5	31.2	25.4	19.9		
		MCWB	14.6	15.6	17.4	19.5	22.4	23.8	24.8	25.1	23.3	22.0	17.8	15.8		
	2%	DB	15.9	18.6	23.4	27.4	30.7	33.6	35.7	34.4	32.7	28.0	22.0	16.7		
		MCWB	12.1	13.3	16.6	18.4	21.7	24.0	25.5	25.1	23.3	19.8	15.4	13.2		
	5%	DB	12.8	15.3	20.8	25.1	28.9	32.4	33.8	32.8	31.0	26.0	19.0	13.6		
		MCWB	9.5	11.1	14.7	17.3	21.0	23.6	25.1	24.4	22.7	18.6	14.3	10.3		
	10%	DB	9.7	12.3	17.9	22.8	27.3	31.0	32.5	31.3	28.7	23.1	17.1	10.9		
		MCWB	6.6	8.5	13.2	16.2	20.4	22.9	24.7	23.7	21.2	17.4	12.9	8.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.6	19.1	21.2	24.6	26.3	27.9	27.6	25.4	22.9	18.8	16.9		
		MCDB	18.5	20.8	24.7	27.3	29.8	32.3	33.1	32.5	31.6	29.0	23.4	19.3		
	2%	WB	12.9	14.4	17.2	19.7	23.3	25.3	26.8	26.4	24.4	21.5	17.0	14.1		
		MCDB	15.0	17.8	21.7	25.1	28.6	31.6	32.7	32.2	30.4	26.2	20.1	16.4		
	5%	WB	9.8	11.6	15.6	18.4	22.1	24.6	26.0	25.5	23.5	20.1	15.2	10.9		
		MCDB	12.8	14.5	19.8	23.3	27.0	30.5	32.0	31.2	28.6	24.0	18.6	12.6		
	10%	WB	6.5	8.3	13.8	17.2	21.3	23.8	25.4	24.6	22.6	18.6	13.1	8.1		
		MCDB	9.5	11.9	17.6	21.7	25.8	29.3	30.9	29.7	27.0	22.5	16.6	10.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	10.9	11.8	11.0	11.2	11.0	11.5	12.8	12.2	10.8	8.8		
		MCDBR	14.2	15.5	14.4	14.8	12.7	12.7	12.3	13.0	14.5	14.8	14.7	12.7		
	5% WB	MCWBR	10.5	10.9	8.7	7.7	5.8	5.0	4.6	5.1	6.0	7.4	9.5	9.5		
		MCWBR	12.9	13.8	12.6	12.3	10.7	11.1	11.0	11.3	11.8	12.0	12.3	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.317	0.350	0.403	0.439	0.449	0.476	0.467	0.411	0.352	0.312	0.303		
	taud		2.472	2.433	2.402	2.249	2.213	2.216	2.143	2.163	2.300	2.468	2.522	2.496		
	Ebn at Noon		873	911	909	874	847	835	808	805	833	858	861	852		
	Edh at Noon		80	95	108	134	142	142	151	145	119	91	76	73		
(44) All-Sky Solar Radiation	RadAvg		2.05	2.76	3.74	4.95	5.56	6.36	6.21	5.58	4.74	3.43	2.32	1.72		
	RadStd		0.23	0.28	0.37	0.28	0.49	0.46	0.43	0.32	0.30	0.37	0.24	0.18		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page